Calculus II

Problems for Training #3 03–07 Mar 2025

◇ **Topics:**

- Partial derivatives
- Chain rule
- Implicit differentiation

In Problems 1–2, find f_x, f_y and f_z.

Problem 1.

$$f(x, y, z) = 1 + xy^2 - 2z^2.$$

Problem 2.

$$f(x, y, z) = e^{-(x^2+y^2+z^2)}.$$

Problem 3. Use the limit definition of partial derivative to compute the partial derivatives of the function at the specified points.

$$f(x, y) = 1 - x + y - 3x^2y, \ \frac{\partial f}{\partial x} \text{ and } \frac{\partial f}{\partial y} \text{ at } (1, 2).$$

Problem 4 Let $w = f(x, y, z)$ be a function of three independent variables. Write down a definition of the partial derivative $\partial f/\partial z$ at (x_0, y_0, z_0) and use it to find $\partial f/\partial z$ at $(1, 2, 3)$ for $f(x, y, z) = x^2yz^2$.

Problem 5. (a) First express dw/dt as a function of t, both by using the Chain Rule and by expressing w in terms of t and differentiating directly with respect to t.

(b) Then evaluate dw/dt at the given value of t.

$$w = \frac{x}{z} + \frac{y}{z}, \quad x = \cos^2 t, \quad y = \sin^2 t, \quad z = 1/t; \quad t = 3.$$

Problem 6. Find $\partial w/\partial v$ when $u = 0$, $v = 0$ if $w = x^2 + (y/x)$, $x = u - 2v + 1$, $y = 2u + v - 2$.

Problem 7. Express $\partial w/\partial r$ and $\partial w/\partial s$ in terms of r and s, if

$$w = x + 2y + z^2, \quad x = r/s, \quad y = r^2 + \ln z, \quad z = 2r.$$

Problem 8. Assuming that the equation below defines y as a differentiable function of x, find the value of dy/dx at the given point.

$$x^3 - 2y^2 + xy = 0, \quad (1, 1).$$

Problem 9. Assuming that the equation below defines y as a differentiable function of x, find the value of dy/dx.

$$y^2 - x^2 - \sin(xy) = 0.$$

Problem 10. Assuming that the equation below defines z as a differentiable function of x and y, find the value of $\partial z/\partial x$ and $\partial z/\partial x$ at $(0, 0, 0)$.

$$x^3 + z^2 + ye^{xz} + z\cos y = 0.$$

END

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Answers to selected problems

3. -13; -2
4. 12
5. (b) 1
6. -7
7. $1/s + 12r; \quad 2/s - r/s^2$
8. $4/3$
10. $0, \quad -1.$